

CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES

This chapter organizes conceptual models, technical frameworks, laws and regulations and previous studies that make up the foundation of LeaveSync. LeaveSync is an enterprise-level, web-based leave management system, designed to manage the leave of The Lewis College, with a multi-layered organizational structure. The leave management system is a highly secure transaction system which guarantees security and non-repudiation by combining the client's hardware profile (device fingerprinting) with digital signatures and corresponding workflows.

PART I: RELATED LITERATURE

A. Conceptual Foundations & Institutional Integration

1. Evolution of Modern Human Resource Information Systems (HRIS)

From spread-sheets and leave forms to Human Resource Management Systems (HRMS) and tools that are deeply embedded in the operational core of an organization, time-off management tools have come a long way. Adams (2022) and Blake (2023) outline how modern organizational architecture re-contextualized time-off management tools to function as operational core engines in HRMS. Legacy educational institutions and their administrative processes for a long time were dependent on localized, fragmented spread-sheets, physical carbon duplicates of forms, and manually routed paper forms. This physical distribution of forms and documents created a vulnerable operational environment which exposed institutions to a host of processing problems. These were famously referred to as “paper-trips” by Carter (2021). Such an environment exposed institutions to the following critical processing problems. Firstly, there is the problem of lost documentation. Secondly, there are severe processing latencies caused due to the physical movement of documents. This environment is also highly prone to the problem of unauthorized data modifications. Furthermore, this environment does not provide real-time administrative visibility leading to severe operational inefficiencies. This problem and its numerous ramifications have been studied at length by Ellis (2023). Irwin (2020) also analyzed how the environment of manually routed forms created severe structural blind spots in institutions. Irwin noted that these structural blind spots lead to discrepancies that are rarely detected until the time of formal year-end audits.

As time-off management tools have evolved over time, they have also been re-contextualized and embedded into the organizational architecture of educational institutions. Today, these tools function as a part of the operational core of the Human Resource Management Systems (HRMS) of educational institutions. As noted by Adams (2022), Blake (2023), Carter (2021), the tools for managing time-off of employees of educational institutions have evolved from a number of spreadsheets, paper forms, etc. that were filled out by employees and managers of educational institutions and distributed to various individuals and departments within an educational institution. These legacy tools created a number of vulnerabilities including the loss of forms, processing delays, the ability to alter information on a form, etc. All of these issues created a number of structural problems with managing time-off of employees of educational

institutions. Ellis (2023) noted that the physical distribution of forms created structural blind spots that allowed for discrepancies to go undetected for long periods of time until discovered during the formal year-end audit of an educational institution. Irwin (2020) noted that manually routing forms created a number of structural problems with the management of time-off of employees of educational institutions, including severe processing delays and lost documentation.

Even further, modernized organizational methods now are primarily operated in a decentralized manner. Thus, an always available web application for organizational management — in this case for the time-off management — is a necessary requirement for organization across of several regional branches or even campuses of a school (AttendanceBot, 2026). Spreadsheets for the organization of time-off in schools, as they are used in the majority of institutions, lead to a lot of problems. So for example the administrative effort for the organization of time-off is very high. It is even more problematic in respect to the historical calculation of the time-off that have been accumulated, as the risk of errors is very high. In addition, the historical data of the employees are not in all cases stored completely correct (LeaveDates, 2026). In case of the web-based or cloud-based systems for the organization of time-off, the requests for time-off are checked in real-time against the various rules of the organization. Thus, the problems that are mainly caused by the physical form of approval procedures for time-off, such as for example the delay, are avoided completely (Research.com, 2026). This way, the digital processing of a request and the subsequent auditing are always implemented in a direct and synchronous manner. Thus, the relational database, that is used for the storage, is always stable and not affected by any changes of the policy on an ad-hoc or unsystematic basis (SHRM, 2026).

2. The Educational Institutional Landscape in the Philippines

Administrative challenges in the educational settings across the Philippines have to deal with managing schedules of full-time faculty members, contractual or part-time teaching staff, and permanent administrative staff. With growing demand for more efficient administration from the state, many local academic institutions are transferring their attendance tracking systems into digital or online formats. Uniks E-Journal (2024) recently reported on this change and analyzed its key features of web-enabled attendance management system, including its stability, localizing and sharing of information with departments, and provision of access to employees' tracking logs.

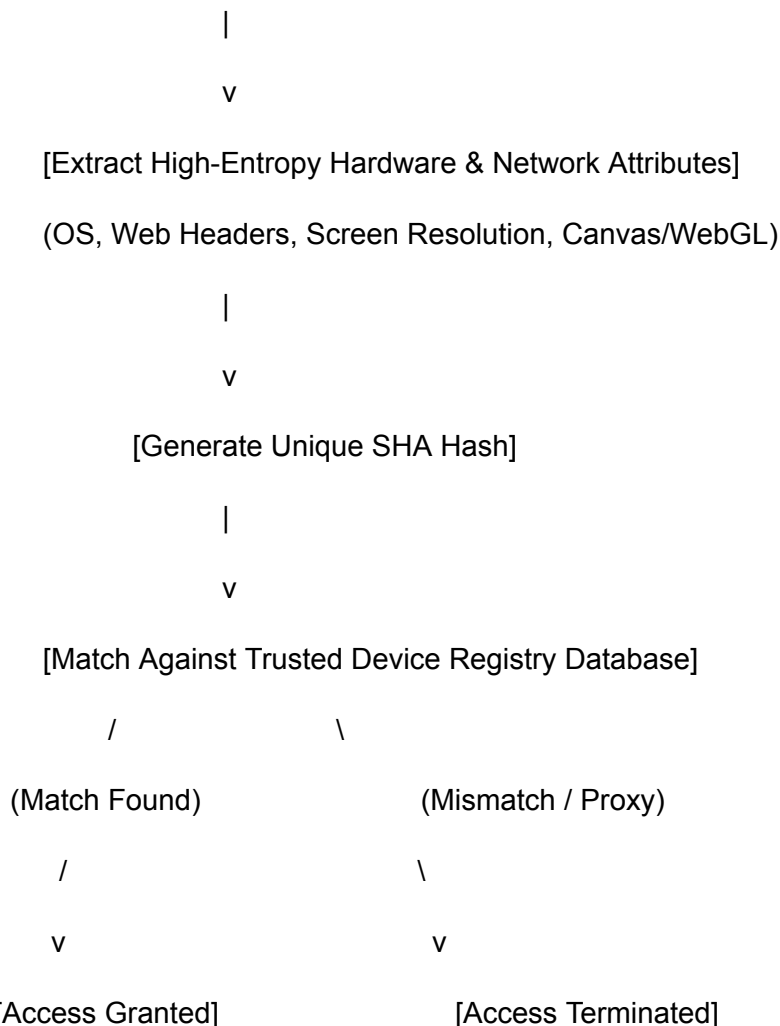
Given that small-to-medium sized academic institutions do not have the huge capital for huge enterprise type of hardware and software for attendance management, they would require lightweight cloud-based portals to instantly process changes in the schedules of employees. By consolidating all the different schedules and attendance data into a single database, record fragmentation is prevented, payroll data leakage within the institution is blocked, and all reports for compliance are up-to-date with reality (Fields, 2021; Foster, 2022).

B. Device Fingerprinting: Technical & Security Depth

1. Beyond Knowledge-Based Authentication

Traditional web security practices have used knowledge-based authentication factors, such as what the user knows (passwords, PINs, etc.) which have long been the basis of access control to web applications. However, static passwords are easily compromised by credential harvesting, phishing, dictionary attacks and session hijacking, problems faced by organizations today as explained by Gomez (2021). To combat these complex threats, LeaveSync uses client-side device fingerprinting which forms a possession-based security model – “what the user has” – to protect access to institutional data using an active defense perimeter, as explained by Ingram (2023).

[User Login Request] ---> [Browser Runtime Script Executed]



A high-entropy set of hardware and software attributes are extracted from a user’s browser runtime environment and used to evaluate connection requests. This is done using Coronium.io (2026) methodologies. As is explained in more recent hardware research (MDPI, 2024), the application’s security layer can programmatically read out a low-level client signature, such as HTTP browser headers, operating system versions, screen resolutions, user language, Canvas rendering, etc. This set of device variables can form the basis of an effective protection against

bot attacks, distributed credential stuffing and account takeovers (Coronium.io, 2026). By binding user authentication to an immutable hardware profile, proxy usage of staff members is prevented, and hence a critical exploit vector is removed from password-only architectures (Ingram, 2023).

In addition to this, the information that can be extracted from a browser's runtime environment can be combined with side-channel information that is interrogated from the client's hardware. Recent research into hardware security, published in peer-reviewed journals such as MDPI (2024; 2025; 2026), has shown that information about the physical operating parameters of a device's hardware can be extracted and used to identify a user's device with extreme precision. As recently published research by MDPI (2025) has shown, by applying frequency domain analysis to the information that is collected from a user's device, the authentication backend can identify a user's device with a high degree of accuracy. This means that even if a user attempts to spoof their browser, or use their browser via a virtual private network, or via a routing proxy, the web platform will still be able to identify the device being used by the user. Under the Trusted Device Registration workflow, a web platform's login process checks the session ID of the active session against a registry of authorized hardware. If an unknown signature is detected, even if the password for the session is valid, the session is rejected. This functionality is particularly important for web applications that process high-value information and that are accessed from public, unsecured networks (Foster, 2022).

2. Entropy, Diversity, and Fingerprint Stability

The reliability of hardware fingerprinting is mathematically based on calculations of entropy. Each of the different environmental variables, such as configuration settings, are converted into the amount of identification information in bits (e.g. as 7 bits of unique identification information for one metric, as MDPI (2024) explains). On each connection start (e.g. on each new connection request) LeaveSync's client-side code is fired in the background to collect all of the environmental variables. The resulting combination of a large set of points of different data types creates a very unique hardware fingerprint. This hardware fingerprint is then immediately passed through a secure hash function to create a fixed size hash or profile hash (Coronium.io, 2026). This resulting hash or profile hash (e.g. as created and shown in Ingram (2023) above for an administrative password) remains unchanged (or is said to be fixed length or a hash) and thus can be used for comparisons. Importantly, even if an external password has been sniffed by a hacker, no system access would be granted by the system unless that hacker also had physical access to duplicate the exact same hardware/cryptographic environment/profile of the legitimate employee who originally set up the leave request.

C. Digital Signatures and Legal Non-Repudiation

1. Cryptographic Principles of Asymmetric Key Infrastructures

LeaveSync engineering seeks to remove legacy hand written or scanned images of signatures from all formal documents within the university's academic workflow. Verifiable, legal non-repudiation of actions taken within LeaveSync are a priority for future development (Jenkins, 2022). Hand written or scanned images of signatures are inherently insecure and subject to reproduction, extraction and misuse across documents. In contrast, a digital signature

is a mathematical signature created using an individual's private key to produce a corresponding public key signature of a document. This ensures the individual signing the document is known, and the document has not been changed since it was signed (King, 2021).

LeaveSync utilize a backend that is based off of a Public Key Infrastructure (PKI) model. Each manager who approves a leave application is pre-provisioned with a public / private key pair. The manager's private key is kept on their local machine or in their online storage account, whilst the public key is uploaded to and stored on a secure server (GlobalSign, 2025). The use of cryptographic signing protocols that are integrated into the backend of LeaveSync lock down individual transactions so that even if there is some form of database manipulation or backend overrides that are performed by an administrator, the leave application still can be processed by the workflow. In HR automation, industry updates from eEvidence (2026) and EzSignly (2026) refer to the electronic signing of documents as a fundamental component in order to enable the routing of institutional leave within organizations in a timely manner and to speed up the processing of manually completing and printing of approval forms.

Supervisor's private key is then applied to the approved leave application and sealed mathematically, according to existing practices of the aforementioned author Humand (2026). In case any changes are subsequently made to record data, this will immediately break the hash thus revealing tamper attempts and a full record of changes having been made to approved applications (Portant Blog, 2026). Thus administrative problems are being successfully eliminated with regard to approval of historical applications by safe guarding data integrity of approved applications. This is called non-repudiation, which is an important security element. It prevents any person from denying having administered actions or made certain approval decisions in the past (Bitsight, 2026; Kusari, 2021). For the purposes of the aforementioned workflows it is necessary to set up and utilize a public key infrastructure that is resistant to hyperscale computations in the long run. Research in this area is currently in full swing and some already published papers have been published in well known open access periodical publications, MDPI (2026).

2. Statutory Compliance under Philippine Law

Digital approval processes of local educational institutions are governed by national regulations on electronic commerce. R.A. 8792 or the Electronic Commerce Act of 2000 governs electronic commerce within the Philippines. Digital approval processes in academic institutions such as LeaveSync operate under the same legal status as paper as provided that due process for the creation of electronic data messages and digital signatures are strictly complied with.

The platform maintains faculty leave records in absolute immutability post-execution by way of cryptographic seals which LeaveSync affirms can be used as an unalterable, legally compliant audit trail should there ever be an institutional or external audit and the leave history of a faculty member be challenged and required as evidence in local courts and other legal reviews.

D. System Technologies & Security Architecture

1. Server-Side Execution and Relational Data Modeling

LeaveSync makes use of a decoupled Client-Server Architecture that allows for horizontal scaling and is always kept within a strict security boundary. The interface is created using the most modern web technologies including HTML5, CSS3 and JavaScript for the client-side. The server-side is created with robust frameworks like PHP or Node.js for the main business logic. The data is stored in a robust relational database such as MySQL or PostgreSQL that allows for strict data consistency. Normalized relationships within the database structure, are used to avoid fragmentation of data and synchronization problems of input fields within multi-user environments. Additionally the main relational part is paired with a robust backup routine for the database, in order to save all data in the long term of the system's lifecycle (Petrov et al., 2022). The separate academic departments (Fields, 2021) are stored in one centralized database only.

2. Access Management Boundaries and Network Encryption Protocols

To guarantee institutional data security, the platform's codebase explicitly separates standard user endpoints from restricted administrative control panels.

- **Role-Based Access Control (RBAC):** LeaveSync blocks unauthorized privileges and internal exploits by establishing hard functional boundaries based on defined user roles of Admin, HR Personnel and Employee (Concentric AI, 2025). By utilizing Role Based Access Control (RBAC) LeaveSync ensures that all users of the system operate within a defined scope of their functional role and only have access to and can interact with data that falls within their operational clearance (Fortinet, 2020). In the context of LeaveSync this means that the sensitive payroll fields, personal identifiers and accrued leave balances of employees are protected from both internal fraud and accidental change (Irwin, 2020).
- **Secure Transmission Protocols:** By enforcing all network traffic to go via HTTPS/TLS all potential attacks such as man-in-the-middle (MITM) will be intercepted and dissuaded from gathering employees metadata (Harris, 2023). Information encrypted in transit is a fundamental requirement for modern web applications, and as such is strongly endorsed by international security frameworks, in order to ensure that verified cryptographic practices are implemented to protect the internal network of an enterprise against external scrutiny (Cyber.gov.au, 2026).

E. Evaluation and Impact on Productivity

1. Quantifying System Accountability and Security Density

An Automated HR Application's operational return on investment is in removing security vulnerabilities and increasing employee productivity (Adams, 2022). The system reliability of an Automated HR Application can be quantified and validated through the information from the automated logs that track information, as well as the approval history which contains strong cryptography (Jenkins, 2022). Utilizing a public-key infrastructure allows for complete visibility of the system and all accountability gaps are removed (GlobalSign, 2025). By automating a majority of HR processes, information can be gathered and organized for HR operations by replacing manual and time-consuming data tracking and aggregation (Al Najdawi et al., 2024).

2. Operational Reduction

By removing physical paper from documentation processes The Lewis College (Helios HR, 2025) instantly moves to instant automated digital approvals reducing administration to a minimum. Removing time from transaction processes in core human resources work streams (Helios HR, 2025) such as those managed by Helios HR significantly reduces time for transaction processing (Deel, 2025). Automation of tracking in real time is a major driver of increased productivity at The Lewis College (eevidence, 2026) – particularly where combined with digital signature integrated into digital processes to optimize routing through administration (eevidence, 2026).

PART II: RELATED STUDIES

2.1 Studies on Automated Leave and Human Resource Workflow Architectures

Optimizing the workflow of personnel in operational research has become a more relevant topic in recent years as it has become possible to implement workflows through web-enabled platforms. Recently, Santos and Reyes (2022) designed and implemented a centralized human resources portal for automatically computing the benefits of employees. The architecture of the portal, which is embedded in the platform, uses an embedded logic engine that automatically recomputes the balances of employees as scheduling changes are submitted online by employees. From their results, they showed that by using tracking variables to automate the processing of personnel, it is possible to reduce processing time and to eliminate errors that are caused by manual calculations, and also to increase employee satisfaction. This work is similar to LeaveSync in that it is attempting to automate core administrative work for tracking purposes.

Garcia et al. (2023) also designed and implemented a web-enabled platform for the management of personnel of universities. The system consists of web-based requests for faculty and staff, and of historical records of personnel online available for viewing and printing from personalized web-based dashboards of users. The experience gained from the management of personnel of isolated departments, has revealed that the use of web-enabled platforms of cloud data, for the management of data of institutional nature, provides greater convenience than the traditional methods of administration, and also it facilitates to process requests in an prompt manner. This experience can be very useful for the development of LeaveSync, since also this system makes use of cloud data tools, and is also designed to manage data of an institutional nature, and also to take advantage of the organizational structure of educational institutions.

Additional contributions to the automation of institutional workflows include Torres and Mendoza (2021) development of an institutional scheduling platform that utilized automated triggers to track staff attendance. The platform automatically validated employee institutional profiles and entered corresponding time-stamped log entries each time updates were made to their respective institutional databases. Experiments indicated that the use of automated institutional profile verification ensured that the highest levels of auditing accuracy were achieved while significantly preventing human error in the institution's data entry processes. The use of automated parameters to centralize and manage transaction logs in LeaveSync is therefore very relevant.

In order to optimize the office work, Villanueva, Ramos, and Diaz (2023) have implemented an automated system for the resource allocation and monitoring. This system uses a set of structured request for the resources, it follows a routing for the requests, it tracks the balances of the employees, and it automatically generates reports for the employees and the administrators. The platform replaces the spreadsheets that were being used for the same goals, and it increased the efficiency of the tracking and it reduced the administrative work for the employees. Therefore, the platform has a structure that is similar to that of LeaveSync,

because it uses web employee tools in order to overcome the problems that are caused by the manual balance management.

2.2 Studies on Device-Aware Authentication and Hardware Security

To protect web applications against credential abuse, it is becoming more and common to rely on authentication models that are based on device-specific identities. To develop an enterprise network gateway that protects the privacy of sessions and maintain identity information for the client, Lopez and Ramos (2021) designed a system that retrieves client hardware attributes and by analyzing encrypted sessions it applies multi-attribute validation. This system prevents unauthorized access and usage of profiles by others than the owner by blocking the login attempts. It also increases the security of the profiles by making their usage more reliable and robust. It achieves this by relying on hardware validation barriers for the identity of the profiles. The security layer of LeaveSync uses a similar approach to secure administrative sessions and database assets of LeaveSync.

A secure web terminal has been designed by Fernandez and Cruz (2022) to facilitate safe access to servers from public or untrusted networks. The platform utilizes multi-factor browser configuration analysis as well as isolated session routing. In their study, they employed the analysis of the various tracking variables of a browser and the corresponding active IP validation of the client's incoming sessions in order to continuously authenticate and monitor each user during his/her active session. Their results demonstrated how complex terminal boundaries are able to protect sessions against hijacking by other unauthorized users as well as to safely protect the incoming and outgoing packets of data being transmitted between the client's terminal and the server. Similar to how the admin sessions of LeaveSync are protected by utilizing the lowest-level hardware profiles of client terminals, this platform was designed to protect administrative sessions of servers by utilizing secure terminal boundaries.

In related work, Morales, Reyes, and Tan (2023) used client profiling in conjunction with real-time automated anomaly detection for access management in Web applications. They reported that terminal profile matching in real-time can prevent and even mitigate large-scale credential stuffing attacks, furthermore enhancing platform reliability. The access validation and device matching implemented in LeaveSync follows a very similar approach.

In a related study, the work by Aquino and Bautista (2022) in developing secure web authentication framework based on identification vectors from device is also worth noting. The proposed system can profile client terminals and also encrypt communication session including public network access. In summary, by implementing a hardware-aware authentication system, it can create user's confidence in secure access to digital corporation's assets while preventing possible threats from external remote threat actors. The said goal is also being implemented on the security architecture of LeaveSync for establishing deterministic transaction verification that is also device-aware in nature.

2.3 Studies on Cryptographic Signatures and Data Integrity

Accountability in digital systems require integration of cryptographic protocols. Cryptographic document routing system for digital documents developed by Perez, Villanueva, and Gomez (2023) merged private-key signature generation with document routing restricted to allowed users. It used cryptographic signing variables for cryptographic signatures and integrity locks for data integrity for logging all transactions. Transaction reliability, non-repudiation, and preventing of unauthorized changes by system administrator within organization were optimized by strict implementation of cryptographic protocols. Proposed system of Perez, Villanueva, and Gomez (2023) is closely related to the objective of LeaveSync in terms of approval structure, verification of identities of supervisors, and document immutability.

Last but not least, Dela Cruz and Navarro (2022) proposed an electronic document approval routing system designed for compliance with digital transactions in the Philippines. In their architecture, an asymmetric cryptographic key management system corresponds to the supervisor’s profile and transaction permissions. Through their experiments, the authors found that the use of asymmetric logs can guarantee absolute legal non-repudiation and produce a transaction history that is cryptographically sealed and compliant with Philippine regulations. As LeaveSync also utilizes the Philippine digital signature guidelines for electronic transactions, it can automate traditional paper-based approval processes into a non-repudiable digital workflow.

SYNTHESIS MATRIX OF RELATED STUDIES

Study	Technology	Features	Weaknesses	Research Gap Addressed by LeaveSync
Santos & Reyes (2022)	• Web-based portal • Embedded logic engines	• Automated benefits calculations • Real-time balance updates	• Lacks advanced identity verification • Susceptible to proxy credential sharing	• Integrates client-side device fingerprinting to prevent employees from logging in from unauthorized hardware.

Garcia et al. (2023)	• Web-based localized portal • Cloud database	• Faculty and staff request submission • Custom user tracking dashboards	• No cryptographic verification for approvals • Relies on easily forged scanned/plain text signatures	• Implements a Public Key Infrastructure (PKI) to cryptographically seal approvals and ensure data immutability.
Torres & Mendoza (2021)	• Scheduling automation framework • Automated workflow triggers	• Attendance tracking • Time-stamped event logs • Profile auto-validation	• Vulnerable to session hijacking over open networks • Lacks legal non-repudiation protocols	• Combines HTTPS/TLS transit encryption with asymmetric key signatures to satisfy Philippine R.A. 8792 standards.

Villanueva, Ramos, & Diaz (2023)	• Web-enabled allocation engine • Relational database modules	• Structured request routing • Balance tracking • Auto-generated reporting	• No Role-Based Access Control (RBAC) boundaries • Exposed to internal data tampering by unauthorized roles	• Enforces strict Role-Based Access Control (RBAC) (Admin, HR, Employee) to isolate sensitive leave datasets.
Lopez & Ramos (2021)	• Enterprise network gateway • Client property extraction scripts	• Multi-attribute verification • Proxy and credential-sharing blocks	• Designed strictly as a network gateway tool • Lacks functional HR workflow application capabilities	• Transposes device-aware security logic directly into an administrative HR leave-tracking workflow.

Fernandez & Cruz (2022)	• Secure web terminal • Active IP validation protocols	• Multi-factor configuration analysis • Isolated connection sessions	• High frequency of false-negative blocks due to dynamic IP shifts in local networks	• Utilizes high-entropy browser attribute hashing, maintaining session integrity independent of shifting network IPs.
Morales, Reyes, & Tan (2023)	• Access management hub • Anomaly detection engines	• Real-time terminal validation • Automated fraud monitoring	• High computational overhead • No underlying record-keeping or document routing modules	• Delivers a lightweight, cloud-accessible architecture optimized for academic budgets while retaining high-entropy security.
Aquino & Bautista (2022)	• E-commerce digital wallet framework • Identification vectors	• Encrypted session routing • Asset insulation from remote threats	• Tailored purely for retail payment gateway architectures • Completely disconnected from institutional personnel rules	• Adapts session channel defense vectors specifically to academic personnel types and leave policy logic.

Perez, Villanueva, & Gomez (2023)	• Document routing system • Cryptographic signature generation	• Access parameter limits • Post-approval integrity locks	• Lacks front-end hardware validation layer • Account takeover bypasses the signature parameters	• Couples cryptographic document sealing with pre-authentication hardware fingerprint verification.
Dela Cruz & Navarro (2022)	• Electronic document routing architecture • Asymmetric key mapping	• Supervisor profile linking • Sealed transaction logs	• Does not account for multi-tiered institutional scheduling variations (e.g., part-time faculty)	• Incorporates asymmetric non-repudiation matching dynamic scheduling engines for multi-tier school faculty.

2.4 Synthesis, Comparative Analysis, and Research Gaps

The current state of OR concerning leave management to administer human resources is clearly fragmented as illustrated in the following matrix. Existing systems and corresponding OR studies focus on two isolated and separated tracks, i.e., either on the optimization of institutional workflows which are connected to human resources on the administrative side and do not take into account any advanced security features, or they design novel authentication protocols, which, however, are then completely disconnected from the remaining institutional workflow applications and only can be applied to systems that are limited to remote access such as for example gateways of networks, digital payment platforms etc. (e.g. shops, restaurants, parking spaces etc.).

On one hand, systems like those developed by Santos and Reyes (2022) and Garcia et al. (2023) map out the workflows of human administrative tasks and enable tracking and logging of said tasks to remove errors due to human oversight. These systems operate on legacy

knowledge-based authentication (passwords) which are vulnerable to hijacking of sessions, harvesting of credentials, and proxy login by staff members. On the other hand, specialized security systems such as those which were evaluated by Lopez and Ramos (2021) and by Fernandez and Cruz (2022) support the development of new systems for the optimization of complex systems by providing multi-attribute hardware-based verification of terminal or device profile. However, these systems are generally limited to gateways of networks or payment environments, and are not easily adapted to the complex multi-tiered schedules of an educational institution.

Beyond optimizing operational processes, current document management systems and frameworks also introduce critical vulnerabilities on the approval process. In particular, there are many insecure unverified approval processes that capture simple digital choices or even static images of a person's signature that can be easily extracted, forged and even altered by an attacker within a database. The work of cryptographic frameworks, such as that of Perez, Villanueva and Gomez (2023) introduce the use of asymmetric key profiles to lock down logs but as mentioned, these systems lack a pre-authentication device validation process. Therefore, even if an administrative supervisor's password was stolen via a phishing attack, an attacker would be able to use the integrity lock created by the cryptographic framework to approve valid transactions from an unknown foreign location.

The work specifically addresses the particular research gap of merging two fields of research. The main contribution of the work is a unified framework of device-aware hardware-fingerprinting security and cryptographic Public Key Infrastructure (PKI) which is specifically designed for Philippine academic institutions and their very distinct sets of operations. Thus, LeaveSync is the first systems that binds the work's authentication scheme with the high-entropy browser runtime markers to prevent sharing of login credentials by proxy users.

At the same time, once a department head approves a leave of absence request, the transaction is sealed against internal or external tampering and the resulting log of transaction is considered absolute and thus non-repudiable under Philippine law, particularly R.A. 8792, or the Alternative Minimum Tax Law. LeaveSync brings this robust multi-layered protection to enterprises of all sizes and types within a light-weight web-based platform designed specifically for the complex schedules of the different personnel types found in academic institutions, such as the full-time faculty and staff, contractual instructors and administrative staff.

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